Exploring the Use of Differential Equations in Electric Circuits

Aneesh Pradhan

Abstract

This paper investigates series RLC circuits, comprising a resistor (R), inductor (L), and capacitor (C) connected in series, through the lens of ordinary differential equations (ODEs). We derive and analyze first-order ODEs governing simplified RC and RL sub-circuits, as well as the second-order ODE describing the full RLC system. Analytic solutions are obtained for all three configurations, and the role of resistance in controlling circuit damping is examined by varying component parameters. Our central research question asks: *How do resistors influence the damping and stability of series RLC circuits as described by ODEs, and what are the practical implications for engineering applications such as filters and oscillators?* The analysis identifies resistance as the parameter that governs whether the circuit is overdamped, critically damped, or underdamped, with direct consequences for signal processing and control system design.

Index Terms

Series RLC circuit, ordinary differential equations, damping, Kirchhoff's Voltage Law, stability analysis, signal processing.

Nomenclature

R	Series resistance, measured in ■.
L	Series inductance, measured in H.
C	Series capacitance, measured in F.
$V(t)$	Applied source voltage.
V_s	Step-source amplitude.
$v_C(t)$	Capacitor voltage, the primary output variable.
$i(t)$	Series current through the resistor, inductor, and capacitor.
α	Damping coefficient, $\alpha = R/(2L)$.
ω_0	Undamped natural frequency, $\omega_0 = 1/\sqrt{LC}$.
ω_d	Damped natural frequency for the underdamped case.
R_{crit}	Critical resistance, $R_{\text{crit}} = 2\sqrt{L/C}$.

I. Introduction

Electric circuits are fundamental to modern technology, powering everything from household appliances to advanced communication systems [1]. Among the most studied configurations is the series RLC circuit, which connects a resistor, an inductor, and a capacitor in series with a voltage source. This topology appears throughout electrical engineering in the design of bandpass filters, resonant oscillators, and impedance-matching networks.

The dynamic behavior of such circuits is naturally captured by *ordinary differential equations* (ODEs). Simpler RC and RL sub-circuits obey *first-order* ODEs, while the complete RLC system requires a *second-order* ODE to describe the interplay between energy stored in the inductor's magnetic field and the capacitor's electric field. Understanding the transition from first- to second-order models clarifies why the resistor plays an indispensable role: it is the only purely dissipative element in the circuit.

A. Pradhan is with the Department of Mathematics, California State University, East Bay, Hayward, CA, USA.
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TABLE I
NOMINAL COMPONENT VALUES

Quantity	Symbol	Nominal value
Source amplitude	V_s	10 V
Resistance cases	R	50 ■, 200 ■, 500 ■
Inductance	L	100 mH = 0.1 H
Capacitance	C	10 µF = $10 \cdot 10^{-6}$ F

Our scientific question is:

How do resistors influence the damping and stability of series RLC circuits as described by ODEs, and what are the practical implications in engineering applications?

The remainder of this paper is structured as follows. Section II introduces the circuit configuration and the governing physical principles. Section III details the derivation of the governing equations, identification of variables and parameters, and the analytic solutions for each circuit type. Section IV presents results from varying circuit parameters and initial conditions. Section V interprets the findings in the context of our research question, discusses limitations, and proposes future work. Section VI concludes the paper.

II. SYSTEM MODEL AND GOVERNING PRINCIPLES

A. Circuit Overview

We consider a series RLC circuit driven by a step voltage source $V(t)$. At $t = 0$, a DC source of amplitude V_s is connected to an initially uncharged capacitor, so $V(t) = V_s$ for the post-switching interval $t \geq 0$. The quantity of primary interest is the capacitor voltage $v_C(t)$ for $t \geq 0$. The component values used for the main RLC case study are summarized in Table I.

These values were chosen because they produce simple, transparent damping parameters while remaining realistic for a bench-top laboratory circuit: $LC = 1 \cdot 10^{-6}\,\mathrm{s}^2$, so $\omega_0 = 1/\sqrt{LC} = 1000\,\mathrm{rad/s}$, and $R_{\mathrm{crit}} = 2\sqrt{L/C} = 200\,\Omega$. Thus $R = 50\,\Omega$, $R = 200\,\Omega$, and $R = 500\,\Omega$ provide clear underdamped, critically damped, and overdamped examples, respectively.

B. State Variables

The two state variables are the capacitor voltage $v_C(t)$ and the inductor current $i(t)$, which also equals the series current through every element. These quantities are chosen because neither can change instantaneously: the capacitor voltage is proportional to stored charge ($v_C = q/C$), and the inductor current is proportional to stored magnetic flux ($\Psi = Li$). Consequently, they fully characterize the energy state of the circuit at any instant.

C. Initial Conditions

Before the switch is closed at $t = 0$:

- The capacitor is initially uncharged: $v_C(0) = 0\,\mathrm{V}$.
- No current flows: $i(0) = 0\,\mathrm{A}$.

Prior to switching, the capacitor behaves as an open circuit (blocking DC current) and the ideal inductor behaves as a short circuit (zero DC voltage drop).

D. Governing Equations

1) Kirchhoff's Voltage Law: Applying Kirchhoff's Voltage Law (KVL) around the series loop [1]:

$$Ri + L\frac{di}{dt} + v_C = V(t). \tag{1}$$

2) Second-Order ODE for the RLC Circuit: Using the capacitor voltage relationship $i(t) = C\,dv_C/dt$ and substituting into (1) yields the governing second-order ODE:

$$LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = V(t). \tag{2}$$

3) First-Order ODEs for Reduced Circuits: For an RC circuit (inductor replaced by an ideal wire, $L \to 0$):

$$\frac{dv_C}{dt} + \frac{1}{RC}\,v_C = \frac{V_s}{RC}. \tag{3}$$

For an RL circuit (capacitor replaced by an ideal wire):

$$\frac{di}{dt} + \frac{R}{L}\,i = \frac{V_s}{L}. \tag{4}$$

III. Analytical Methods

A. Equations, Conditions, and Connection to the Application

The three ODEs derived above–(3), (4), and (2)–together model the spectrum from simple to complex circuit behavior. With initial conditions applied, the initial value problems (IVPs) are:

RC circuit (first-order):

$$\frac{dv_C}{dt} + \frac{1}{RC}\,v_C = \frac{V_s}{RC}, \qquad v_C(0) = V_0. \tag{5}$$

RL circuit (first-order):

$$\frac{di}{dt} + \frac{R}{L}\,i = \frac{V_s}{L}, \qquad i(0) = 0. \tag{6}$$

RLC circuit (second-order):

$$LC\frac{d^2v_C}{dt^2} + RC\frac{dv_C}{dt} + v_C = V_s, \qquad v_C(0) = 0\,\mathrm{V}, \quad \dot{v_C}(0) = 0. \tag{7}$$

Each equation models a physical balance: the left-hand terms represent energy dissipation (via R) and energy storage (via L and C), while the right-hand side represents the forcing from the voltage source.

B. Independent and Dependent Variables

- **Independent variable:** time t in s.
- **Dependent variables:** $v_C(t)$ in V (capacitor voltage) and $i(t)$ in A (series current).

C. Parameters

- R: resistance; dissipates electrical energy as heat. Appears in the first-order damping term R/L and the second-order damping coefficient $\alpha = R/(2L)$.
- $L = 0.1\,\mathrm{H}$: inductance; stores energy in a magnetic field. Governs how quickly current can change.
- $C = 10 \cdot 10^{-6}\,\mathrm{F}$: capacitance; stores energy in an electric field. Governs how quickly voltage can change.
- $V_s = 10\,\mathrm{V}$: step-input amplitude. Represents a DC voltage switched on at $t = 0$.

D. Assumptions and Limitations

a) Assumptions.: We adopt the standard *lumped-element model*: all resistance is concentrated in R, all inductance in L, and all capacitance in C, with no parasitic coupling between components. Component values are assumed to be constant with respect to temperature, frequency, and signal amplitude (i.e., linear, time-invariant behavior). Wires and switch contacts are treated as ideal (zero resistance, zero inductance).

b) Limitations.: Real resistors exhibit temperature-dependent resistance governed by $R(T) = R_0[1 + \alpha_R(T - T_0)]$, introducing nonlinear behavior under large current loads. At high frequencies, parasitic inductance in capacitors and parasitic capacitance in inductors violate the lumped-element assumption, requiring distributed-element or transmission-line models. The ideal step-input assumption ignores source impedance and switching transients present in physical systems. These effects are left for future work.

E. Analytic Solutions

1) First-Order RC Circuit: Equation (5) is a linear first-order ODE with constant coefficients. Using the integrating factor $\mu(t) = e^{t/(RC)}$, the general solution is:

$$v_C(t) = V_s + (V_0 - V_s)\, e^{-t/(RC)}. \tag{8}$$

The time constant $\tau_{RC} = RC$ determines how quickly v_C relaxes to the steady-state value V_s.

2) First-Order RL Circuit: Similarly, (6) yields:

$$i(t) = \frac{V_s}{R}\left(1 - e^{-(R/L)t}\right). \tag{9}$$

The time constant is $\tau_{RL} = L/R$.

3) Second-Order RLC Circuit: Characteristic Equation and Damping: Seeking homogeneous solutions of the form e^{rt} in (7) leads to the characteristic equation:

$$LCr^2 + RCr + 1 = 0. \tag{10}$$

Dividing by LC and introducing the standard parameters

$$\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \tag{11}$$

the roots become $r = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$. The discriminant $\Delta = \alpha^2 - \omega_0^2$ determines damping behavior:

- **Overdamped** ($\Delta > 0$, i.e. $R > 2\sqrt{L/C}$): two distinct real negative roots. The circuit returns to equilibrium without oscillating, but slowly. This occurs in high-resistance configurations such as lossy filters.
- **Critically damped** ($\Delta = 0$, i.e. $R = 2\sqrt{L/C}$): one repeated real root $r = -\alpha$. Fastest non-oscillatory return to equilibrium; used in precision instruments that must settle quickly.
- **Underdamped** ($\Delta < 0$, i.e. $R < 2\sqrt{L/C}$): two complex conjugate roots $r = -\alpha \pm j\omega_d$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ is the damped natural frequency. The circuit oscillates with exponentially decaying amplitude; typical in resonant and tuned circuits.

4) Second-Order RLC Circuit: General Analytic Solution:

a) Homogeneous solutions.: For each damping case, the general homogeneous solution is:
Overdamped (r_1, r_2 real, distinct):

$$v_{C,h}(t) = A_1 e^{r_1 t} + A_2 e^{r_2 t}. \tag{12}$$

Critically damped ($r = -\alpha$ repeated):

$$v_{C,h}(t) = (A_1 + A_2 t)\, e^{-\alpha t}. \tag{13}$$

Underdamped (complex roots):

$$v_{C,h}(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)\,. \tag{14}$$

b) Non-homogeneous solution.: For a constant forcing $V(t) = V_s$, the particular solution is simply $v_{C,p} = V_s$. For time-varying inputs, two systematic methods apply: (i) *Variation of parameters*, which replaces the constants A_1, A_2 with functions $u_1(t), u_2(t)$ determined by a two-equation system derived from the ODE; or (ii) *Laplace transforms*, which convert the ODE to an algebraic equation $\mathcal{L}\{v_C\}(s) = \mathcal{L}\{V(t)\}(s) \cdot H(s)$, where $H(s) = 1/(LCs^2 + RCs + 1)$ is the circuit transfer function, followed by a partial-fraction expansion and inverse transform.

TABLE II
DAMPING CASES FOR THE SERIES RLC STEP RESPONSE

Case	R	α	Characteristic data
Underdamped	50 ■	250/s	$\omega_d = 968.25\,\mathrm{rad/s}$
Critically damped	200 ■	1000/s	$r = -1000/\mathrm{s}$
Overdamped	500 ■	2500/s	$r_1 = -208.71/\mathrm{s}$, $r_2 = -4791.29/\mathrm{s}$

c) Applying initial conditions (underdamped case).: The complete solution is $v_C(t) = v_{C,h}(t) + V_s$. Applying $v_C(0) = V_0$ and $\dot{v_C}(0) = 0$:

$$v_C(0) = A_1 + V_s = V_0 \implies A_1 = V_0 - V_s, \tag{15}$$

$$\dot{v_C}(0) = -\alpha A_1 + \omega_d A_2 = 0 \implies A_2 = \frac{\alpha(V_0 - V_s)}{\omega_d}. \tag{16}$$

Substituting (15)–(16) into (14) and adding V_s gives the complete analytic solution for the underdamped case.

IV. RESULTS

A. First-Order RC Response

For the RC sub-circuit, the governing equation can be written as

$$\dot{v_C} = \frac{V_s - v_C}{RC}. \tag{17}$$

The equilibrium is $v_C^* = V_s = 10\,\mathrm{V}$. If $v_C < V_s$, then $\dot{v_C} > 0$; if $v_C > V_s$, then $\dot{v_C} < 0$. Thus the phase line points toward v_C^* from both sides, so the equilibrium is asymptotically stable.

Using $R = 100\,\Omega$ and $C = 10\,\mu\mathrm{F}$ gives $\tau_{RC} = RC = 1\,\mathrm{ms}$. For three initial capacitor voltages,

$$v_C(t; V_0 = 0) = 10\left(1 - e^{-1000t}\right), \tag{18}$$

$$v_C(t; V_0 = 5) = 10 - 5e^{-1000t}, \tag{19}$$

$$v_C(t; V_0 = 10) = 10. \tag{20}$$

All three cases approach the same stable equilibrium, but they start from different points on the phase line.

With $V_0 = 0$ fixed and $C = 10\,\mu\mathrm{F}$, varying resistance changes only the time constant:

$$R = 50\,\Omega: \quad v_C(t) = 10\left(1 - e^{-2000t}\right), \tag{21}$$

$$R = 100\,\Omega: \quad v_C(t) = 10\left(1 - e^{-1000t}\right), \tag{22}$$

$$R = 200\,\Omega: \quad v_C(t) = 10\left(1 - e^{-500t}\right). \tag{23}$$

Larger resistance produces slower charging, which is the first-order analog of the damping effect studied in the full RLC circuit.

B. Second-Order RLC Response

For the selected component values,

$$\omega_0 = 1000\,\mathrm{rad/s}, \qquad R_{\mathrm{crit}} = 200\,\Omega. \tag{24}$$

The three resistance values in Table II therefore sample the underdamped, critically damped, and overdamped regimes without changing L, C, or the source voltage.

With $v_C(0) = 0$, $\dot{v_C}(0) = 0$, and $V_s = 10\,\mathrm{V}$, the numerical step responses are as follows, with t measured in seconds and v_C measured in volts:

$$v_{C,50}(t) = 10\big[1 - e^{-250t}(\cos(968.25t) + 0.2582\sin(968.25t))\big]\,, \tag{25}$$

$$v_{C,200}(t) = 10\big[1 - (1 + 1000t)e^{-1000t}\big]\,, \tag{26}$$

$$v_{C,500}(t) = 10 - 10.456e^{-208.71t} + 0.456e^{-4791.29t}. \tag{27}$$

Equation (25) oscillates around the final value before settling, (26) reaches the final value as quickly as possible without oscillation, and (27) approaches the final value without overshoot but is slowed by the dominant pole $r_1 = -208.71/\mathrm{s}$.

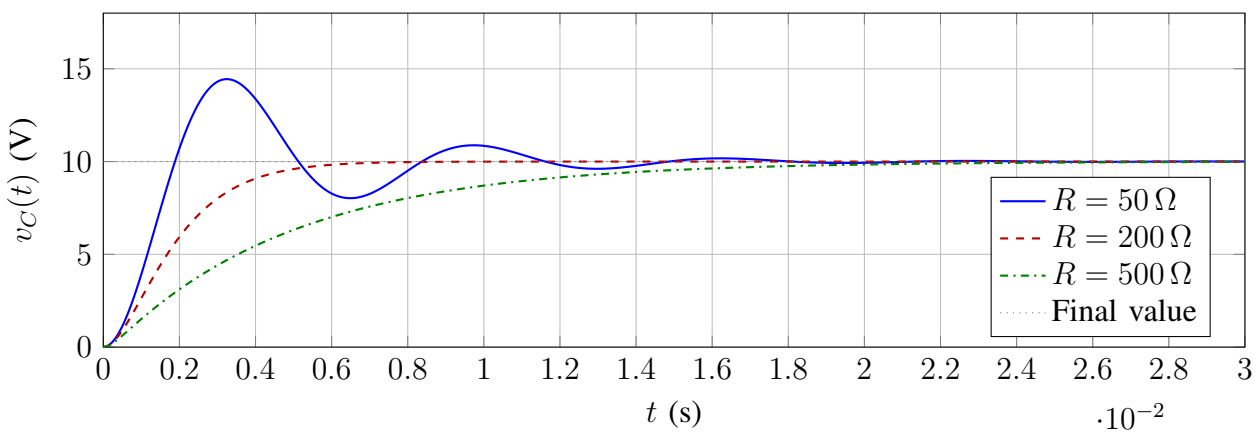

Fig. 1. Capacitor voltage $v_C(t)$ for three values of resistance R, showing underdamped, critically damped, and overdamped responses.

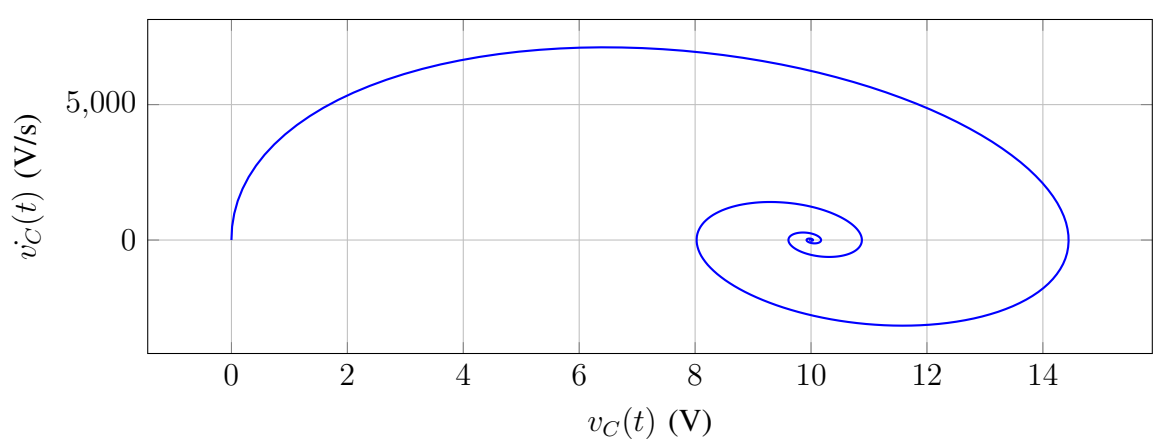

Fig. 2. Phase portrait of the second-order system ($\dot{v_C}$ vs. v_C) for the underdamped case with $R = 50\,\Omega$.

C. Connection to the Research Question

The results show that changing R changes the characteristic roots while leaving the energy-storage parameters L and C fixed. For $R = 50\,\Omega$, the roots are complex and the capacitor voltage rings before settling. For $R = 200\,\Omega$, the repeated real root gives the fastest non-oscillatory response. For $R = 500\,\Omega$, two negative real roots remove oscillation but slow the return to equilibrium. Thus resistance directly controls the damping mechanism described in the research question.

V. Discussion

A. Summary of Problem and Results

This paper modeled series RLC circuits using first- and second-order ODEs, deriving analytic solutions for RC, RL, and RLC configurations. The resistor was identified as the sole energy-dissipating element, and its value shown to directly determine the discriminant of the characteristic equation (10), thereby controlling whether the circuit is overdamped, critically damped, or underdamped.

B. Interpretation in Terms of the Research Question

Without resistance ($R = 0$), the characteristic roots are purely imaginary ($r = \pm j\omega_0$), yielding perpetual, undamped oscillations in an ideal LC tank circuit. Any nonzero R introduces the decay envelope $e^{-\alpha t}$ that eventually damps oscillations to zero. In engineering applications: bandpass filters rely on controlled underdamping to achieve frequency selectivity when the output is taken across the resistor, while the capacitor-voltage response studied here behaves like a low-pass response with possible resonant overshoot. Overdamped circuits ensure no ringing in power supplies and ADC front-ends; critically damped designs appear in servo systems where settling speed is paramount. A passive RLC circuit can store and exchange energy, but a sustained oscillator also requires an active energy source or feedback element to replace resistive losses. This analytical framework directly addresses the research question: resistors are essential for stabilizing oscillatory behavior and enabling practical circuit design.

C. Limitations and Future Work

The lumped-element and linearity assumptions restrict the model's validity to low frequencies and small signals. Future work should incorporate: (i) a nonlinear resistance model $R(T, i)$ to capture self-heating; (ii) distributed-element analysis for RF applications above 1 MHz; (iii) experimental validation using an oscilloscope to compare measured $v_C(t)$ waveforms against the analytic solutions derived here; and (iv) extension to parallel RLC and more complex multi-mesh circuits modeled by systems of ODEs.

VI. Conclusion

We derived and solved first- and second-order ODEs governing RC, RL, and series RLC circuits, demonstrating analytically and qualitatively how the resistor controls damping. The three damping regimes– overdamped, critically damped, and underdamped– correspond directly to engineering trade-offs between settling speed and oscillatory behavior. The analysis shows that resistors are not merely optional loss elements but are indispensable for stable, predictable circuit operation.

References

[1] J. W. Nilsson and S. A. Riedel, *Electric Circuits*, 12th ed. Hoboken, NJ: Pearson, 2019.